FUNCTIONS OF RANDOM VARIABLES

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1. Functions of Random Variables
   - Why Functions of Random Variables Are Important?
   - A Common Misperception on Functions of Random Variables

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2. Functions of Discrete Random Variables
   - Expected Value of A Function of A DRV
   - Variance of A Function of A DRV
   - Distribution of A Function of A DRV

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   - Why Functions of Random Variables Are Important?
   - A Common Misperception on Functions of Random Variables
2. Functions of Discrete Random Variables
   - Expected Value of A Function of A DRV
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Functions of Random Variables

- We have said that, a random variable is a real-valued representation of an outcome in the sample space.
  - X: Number of defective products in a production batch is a discrete random variable.
  - Y: Daily demand to gasoline in a gas station is a continuous random variable.
- We are often interested in measuring the benefit (or loss) caused by a random occurence. This benefit/loss, sometimes, can be modeled as a mathematical function of the RV of interest.
  - $g(X)$: The amount of monetary loss caused by the defective products.
  - $h(Y)$: Daily profit obtained through the daily demand to gasoline.
- While X represents the random input (resource) of a system, $g(X)$ may represent the output (performance) of the system.

Functions of Random Variables

- Given a random variable X and a real valued function
$$g : \mathbb{R} \to \mathbb{R}$$
the transformation $g(X)$ is also a random variable in most of the cases.
  - If the daily demand is random, daily profit is also random, based on the randomness of the demand.
- Therefore, $g(X)$ has its:
  - own expectation,
  - own variance,
  - own distribution functions (CDF and PMF/PDF).
- How do we determine the distribution of $g(X)$?

A Common Misperception on Functions of RVs

- Suppose X is a RV and $g(X)$ is a real-valued function.
- The expected value of $g(X)$ is denoted by $E[g(X)]$.
  - In most cases:
$$E[g(X)] \neq g(E[X])$$
  - Proving example: $E\left[X^2\right] \neq E[X]^2$.
  - A counter example: If g is a linear function, the equation holds.
$$E[aX+b] = aE[X]+b$$
- The variance of $g(X)$ is denoted by $Var[g(X)]$.
  - In most cases:
$$Var[g(X)] \neq g(Var[X])$$
  - Proving example: $Var[aX+b] \neq aVar[X]+b$.
- The distribution of $g(X)$ is also different.

Expected Value of A Function of A DRV

- Suppose X is a **discrete** RV along with its PMF, $p(x)$.
- Suppose further, $g(X)$ is a real-valued function.
- The expected value of $g(X)$, is computed by:

$$E[g(X)] = \sum_{x:p(x)>0} g(x)\,p(x)$$

Expected Value of A Function of A DRV: Example

Q1. Let X be a random variable with PMF:

$$p(x) = \begin{cases} \frac{x+3}{15}, & x = -2,-1,0,1,2 \\ 0, & \text{elsewhere} \end{cases}$$

Compute:

a. $E[|X|]$
b. $E\left[e^X\right]$
c. $E\left[\max\left\{X^2-3,0\right\}\right]$

A1. X is a discrete random variable. Hence, we will use:

$$E[g(X)] = \sum_{x:p(x)>0} g(x)\,p(x)$$

where g is defined differently in (a), (b), and (c).

Expected Value of A Function of A DRV: Example (cont'd)

a. $g(x) = |x|$

$$E[|X|] = \sum_{x=-2}^{2} \frac{|x|(x+3)}{15} = |-2|\frac{1}{15} + |-1|\frac{2}{15} + |0|\frac{3}{15} + |1|\frac{4}{15} + |2|\frac{5}{15} = 1.2$$

b. $g(x) = e^x$

$$E\left[e^X\right] = \sum_{x=-2}^{2} \frac{e^x(x+3)}{15} = \frac{e^{-2}}{15} + \frac{2e^{-1}}{15} + \frac{3e^0}{15} + \frac{4e^1}{15} + \frac{5e^2}{15} \approx 3.446$$

c. $g(x) = \max\left\{x^2 - 3, 0\right\}$

$$E\left[\max\left\{X^2 - 3, 0\right\}\right] = \sum_{x=-2}^{2} \frac{\max\left\{x^2-3,0\right\}(x+3)}{15} = (1)\frac{1}{15} + (0)\frac{2}{15} + (0)\frac{3}{15} + (0)\frac{4}{15} + (1)\frac{5}{15} = 0.4$$

Expected Value of A Function of A DRV: Example II

Q2. In a market research procedure, the test is performed over groups of at most 20 people and the test returns more reliable results if the group size is larger. The researcher invites 20 guests to a session and, independent of each other, each guest is likely to participate with probability 0.2. After the test is performed, the researcher will reward the participants with a total amount of money.

- If less than 3 people participate, every participant will be paid 1₺.
- Otherwise, every participant will be paid x₺, where x is the number of participants in the session.

What is the expected amount that the researcher needs to pay in any session?

Expected Value of A Function of A DRV: Example II (cont'd)

A2. Let X be the number of participants in a session, that is:

$$X \sim Binom\left(n = 20, p = 0.2\right)$$

and, whenever x people participate, the total amount that the researcher needs to pay in a session as a function of x is:

$$g\left(x\right) = \begin{cases} x, & x < 3 \\ x^2, & x \geq 3 \end{cases}$$

The expected amount that the researcher needs to pay in any session is $E\left[g\left(X\right)\right]$.

Expected Value of A Function of A DRV: Example II (cont'd)

A2. $X \sim Binom(n = 20, p = 0.2)$

$$\begin{aligned} E[g(X)] &= \sum_{x=0}^{2} xp(x) + \sum_{x=3}^{20} x^2 p(x) \\ &= \sum_{x=0}^{2} \left(x - x^2\right) p(x) + \sum_{x=0}^{20} x^2 p(x) \\ &= -2p(2) + E\left[X^2\right] \\ &= -2p(2) + Var[X] + E[X]^2 \approx 18.9262 \end{aligned}$$

since:

$$Var[X] = np(1-p) = 20(0.2)(0.8) = 3.2$$
$$E[X]^2 = (np)^2 = (20 \times 0.2)^2 = 16$$
$$p(2) = P\{X = 2\} = \binom{20}{2}(0.2)^2(0.8)^{18} \approx 0.1369$$

Variance of A Function of A DRV

- Suppose X is a **discrete** RV along with its PMF, $p(x)$.
- Suppose further, $g(X)$ is a real-valued function.
- The variance of $g(X)$, is computed by:

$$\begin{aligned} Var[g(X)] &= E\left[g^2(X)\right] - E[g(X)]^2 \\ &= \sum_{x:p(x)>0} g^2(x)\,p(x) - \left(\sum_{x:p(x)>0} g(x)\,p(x)\right)^2 \end{aligned}$$

Variance of A Function of A DRV: Example

Q3. Consider the random variable X in Question 1. Remember its PMF:

$$p(x) = \begin{cases} \frac{x+3}{15}, & x = -2, -1, 0, 1, 2 \\ 0, & \text{elsewhere} \end{cases}$$

Compute $Var[|X|]$.

Variance of A Function of A DRV: Example

Q3. Consider the random variable X in Question 1. Remember its PMF:

$$p(x) = \begin{cases} \frac{x+3}{15}, & x = -2,-1,0,1,2 \\ 0, & \text{elsewhere} \end{cases}$$

Compute $Var[|X|]$.

A3. We know $E[|X|] = 1.2$. Let us compute $E\left[|X|^2\right]$.

$$E\left[|X|^2\right] = \sum_{x=-2}^{2} \frac{|x|^2(x+3)}{15} = |-2|^2 \frac{1}{15} + |-1|^2 \frac{2}{15} + |0|^2 \frac{3}{15} + |1|^2 \frac{4}{15} + |2|^2 \frac{5}{15} = 2$$

$$Var[|X|] = E\left[|X|^2\right] - E[|X|]^2 = 2 - (1.2)^2 = 0.56$$

Finding PMF For A Function of A DRV

- Suppose X is a **discrete** RV along with its PMF, $p_X(x)$.
- Suppose further, $Y = g(X)$ is a real-valued function of X.
- In order to identify the distribution of Y, we need to determine its PMF, $p_Y(y)$.
- The procedure is pretty straightforward:
  - Identify the set of values $\{x_1, x_2, \ldots\}$ that X can take.
  - For all x values in the above set, compute $y = g(x)$ and add probability $p_X(x)$ to $p_Y(y)$.

Finding PMF For A Function of A DRV: Example

Q4. Let X be a random variable with PMF:

$$p_X(x) = \begin{cases} \frac{x+3}{15}, & x = -2,-1,0,1,2 \\ 0, & \text{elsewhere} \end{cases}$$

Let $Y = \max\left\{1, \sqrt{X^2}\right\}$ be a new random variable. Compute the PMF of Y.

A4. X can take $\{-2,-1,0,1,2\}$ values according to probabilities given in the PMF. For each of these values, we will compute the respective value that Y can take and add $P\{X = x\}$ to that value in the PMF of Y.

Finding PMF For A Function of A DRV: Example (cont'd)

A4. Check the following tabulation for the answer.

x	$y = \max\left\{1, \sqrt{x^2}\right\}$	$p_X(x)$
-2	2	$1/15$
-1	1	$2/15$
0	1	$3/15$
1	1	$4/15$
2	2	$5/15$

y	$p_Y(y)$
1	$\frac{2}{15} + \frac{3}{15} + \frac{4}{15} = \frac{9}{15}$
2	$\frac{1}{15} + \frac{5}{15} = \frac{6}{15}$

Hence:

$$p_Y(y) = \begin{cases} 0.6, & y = 1 \\ 0.4, & y = 2 \\ 0, & \text{elsewhere} \end{cases}$$

Expected Value of A Function of A CRV

- Suppose X is a **continuous** RV along with its PDF, $f(x)$.
- Suppose further, $g(X)$ is a real-valued function.
- The expected value of $g(X)$, is computed by:

$$E[g(X)] = \int_{-\infty}^{\infty} g(x) f(x) dx$$

Expected Value of A Function of A CRV: Example

Q5. Let X be continuous standard uniform random variable. We define three new random variables $Y = X^2$, $W = X^3$, $T = X^4$. Calculate $E[Y]$, $E[W]$, and $E[T]$.

Expected Value of A Function of A CRV: Example

Q5. Let X be continuous standard uniform random variable. We define three new random variables $Y = X^2$, $W = X^3$, $T = X^4$. Calculate $E[Y]$, $E[W]$, and $E[T]$.

A5. Let us compute $E[X^n]$ for $n \in \mathbb{Z}^+$:

$$X \sim Unif\{\alpha = 0, \beta = 1\} \Rightarrow f_X(x) = \begin{cases} 1, & 0 \leq x \leq 1 \\ 0, & \text{elsewhere} \end{cases}$$

$$E[X^n] = \int_0^1 x^n f(x)\,dx = \int_0^1 x^n dx = \frac{x^{n+1}}{n+1}\Big|_0^1 = \frac{1}{n+1}$$

Hence:

$$E[Y] = \frac{1}{3}, \quad E[W] = \frac{1}{4}, \quad E[T] = \frac{1}{5}$$

Expected Value of A Function of A CRV: Example II

Q6. Let the probability density function of X be given by:

$$f(x) = \begin{cases} \frac{32}{(x+4)^3} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

Compute $E[X]$.

Expected Value of A Function of A CRV: Example II

Q6. Let the probability density function of X be given by:

$$f(x) = \begin{cases} \frac{32}{(x+4)^3} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

Compute $E[X]$.

A6. It is more convenient to compute $E[X+4]$ at first.

$$E[X+4] = \int_0^\infty (x+4) \frac{32}{(x+4)^3} dx = 32 \int_0^\infty (x+4)^{-2} dx$$

If we substitute $y = x+4$ and $dy = dx$, we get:

$$32 \int_0^\infty (x+4)^{-2} dx = 32 \int_4^\infty y^{-2} dy = -\frac{32}{y}\Big|_4^\infty = 0 - (-8) = 8$$

Finally, $E[X+4] = E[X] + 4$. Hence, $E[X] = 4$

Expected Value of A Function of A CRV: Example III

Q7. Let the probability density function of X be given by:

$$f(x) = \begin{cases} 6x(1-x) & x \in [0,1] \\ 0 & \text{otherwise} \end{cases}$$

Compute $E\left[X^3\right]$.

Expected Value of A Function of A CRV: Example III

Q7. Let the probability density function of X be given by:

$$f(x) = \begin{cases} 6x(1-x) & x \in [0,1] \\ 0 & \text{otherwise} \end{cases}$$

Compute $E\left[X^3\right]$.

A7. $E\left[X^3\right] = \int_0^1 x^3 6x(1-x)\,dx = 6\int_0^1 x^4 - x^5 dx = \left[6\frac{x^5}{5} - x^6\right]_0^1 = \frac{1}{5}$

Variance of A Function of A CRV

- Suppose X is a **continuous** RV along with its PDF, $f(x)$.
- Suppose further, $g(X)$ is a real-valued function.
- The variance of $g(X)$, is computed by:

$$\begin{aligned} Var[g(X)] &= E\left[g^2(X)\right] - E[g(X)]^2 \\ &= \int_{-\infty}^{\infty} g^2(x) f(x)\, dx - \left(\int_{-\infty}^{\infty} g(x) f(x)\, dx\right)^2 \end{aligned}$$

Variance of A Function of A CRV: Example

Q8. Let X be continuous standard uniform random variable. Compute the variance of $Y = X^n$ for $n \in \mathbb{Z}^+$.

Variance of A Function of A CRV: Example

Q8. Let X be continuous standard uniform random variable. Compute the variance of $Y = X^n$ for $n \in \mathbb{Z}^+$.

A8. Note that:

$$Var\left[X^n\right] = E\left[X^{2n}\right] - E\left[X^n\right]^2$$

Also $X \sim Unif\left(\alpha = 0, \beta = 1\right)$ and we know that $E\left[X^n\right] = \frac{1}{n+1}$.
Hence:

$$\begin{aligned} Var\left[X^n\right] &= E\left[X^{2n}\right] - E[X^n]^2 = \frac{1}{2n+1} - \left(\frac{1}{n+1}\right)^2 \\ &= \frac{1}{2n+1} - \frac{1}{n^2+2n+1} = \frac{n^2}{(2n+1)(n+1)^2} \end{aligned}$$

Identifying the Distribution of A Function of A CRV

- Suppose X is a **continuous** RV along with its PDF, $f_X(x)$.
- Suppose further, $Y = g(X)$ is a real-valued function of X.
- In order to identify the distribution of Y, we need to determine its CDF $F_Y(y)$ using the CDF of X, $F_X(x)$.
  - First, determine the values that Y can take.
  - Then, compute the CDF $F_X(x)$ of X.
  - If the function g is a strictly monotonic[1] and differentiable function, the procedure is straightforward.
  - Otherwise, there is no specified procedure. Finding the distribution of Y may or may not be practical.
- If we can compute $F_Y(y)$, we can also compute the PDF of Y:

$$f_Y(y) = \frac{\partial F_Y(y)}{\partial y}$$

[1] always increasing or always decreasing

Identifying the Distribution of A Function of A CRV (cont'd)

- **Theorem of Transformation:** When the function g is a strictly monotonic and differentiable function, then the PDF of $Y = g(X)$ is:

$$f_Y(y) = f_X\left[g^{-1}(y)\right]\left|\frac{\partial}{\partial y}g^{-1}(y)\right|$$

Proof: Suppose g is strictly increasing[2]:

$$\begin{aligned} F_Y(y) &= P(Y \leq y) = P(g(X) \leq y) \\ &= P\left(g^{-1}[g(X)] \leq g^{-1}(y)\right) \\ &= P\left(X \leq g^{-1}(y)\right) \\ &= F_X\left[g^{-1}(y)\right] \end{aligned}$$

$$f_Y(y) = \frac{\partial F_Y(y)}{\partial y} = \frac{\partial}{\partial y}F_X\left[g^{-1}(y)\right] = f_X\left[g^{-1}(y)\right]\frac{\partial}{\partial y}g^{-1}(y)$$

[2] For strictly decreasing g, the absolute value function is put in use.

Identifying the Dist. of A Function of A CRV: Example

Q9. Let X be a continuous random variable with the following PDF:

$$f_X(x) = 2x, \quad x \in [0, 1]$$

We define random variable $Y = \sqrt{X}$ as a function of X. Compute the PDF $f_Y(y)$ of Y.

Identifying the Dist. of A Function of A CRV: Example

Q9. Let X be a continuous random variable with the following PDF:

$$f_X(x) = 2x, \quad x \in [0, 1]$$

We define random variable $Y = \sqrt{X}$ as a function of X. Compute the PDF $f_Y(y)$ of Y.

A9. **Step 1:** Note that Y can also take values between 0 and 1.

Identifying the Dist. of A Function of A CRV: Example

Q9. Let X be a continuous random variable with the following PDF:

$$f_X(x) = 2x, \quad x \in [0,1]$$

We define random variable $Y = \sqrt{X}$ as a function of X. Compute the PDF $f_Y(y)$ of Y.

A9. **Step 1:** Note that Y can also take values between 0 and 1.

Step 2: Determine the CDF of X.

$$F_X(x) = \int_{-\infty}^{x} f_X(t)\,dt = \int_{0}^{x} 2t\,dt = t^2\Big|_0^x = x^2, \quad x \in [0,1]$$

Identifying the Dist. of A Func. of A CRV: Example (cont'd)

Step 3: In order to determine the PDF of Y, first we need to determine its CDF $F_Y(y)$.

$$F_Y(y) = P\{Y \leq y\} = P\left\{\sqrt{X} \leq y\right\} = P\left\{0 \leq X \leq y^2\right\}$$

Since X can not be negative:

$$F_Y(y) = P\left\{X \leq y^2\right\} = F_X\left(y^2\right) = \left(y^2\right)^2 = y^4, \quad y \in [0,1]$$

or in a more detailed form:

$$F_Y(y) = \begin{cases} 0, & y < 0 \\ y^4, & 0 \leq y \leq 1 \\ 1, & y > 1 \end{cases}$$

Identifying the Dist. of A Func. of A CRV: Example (cont'd)

Step 4: The PDF of Y can be computed by:

$$f_Y(y) = \frac{\partial F_Y(y)}{\partial y} = 4y^3, \quad y \in [0,1]$$

or in a more detailed form:

$$F_Y(y) = \begin{cases} 0, & y < 0 \\ y^4, & 0 \leq y \leq 1 \\ 1, & y > 1 \end{cases} \Rightarrow f_Y(y) = \begin{cases} 4y^3, & 0 \leq y \leq 1 \\ 0, & \text{elsewhere} \end{cases}$$

- This result could have been directly obtained using the theorem of transformation, since $g(x) = \sqrt{x}$ is a strictly monotonic and differentiable in $[0,1]$.

Identifying the Dist. of A Function of A CRV: Example II

Q10. Let $X \sim Unif\left(\alpha = -2, \beta = 2\right)$: We define random variable $Y = |X|$ as a function of X. Compute the PDF $f_Y\left(y\right)$ of Y.

Identifying the Dist. of A Function of A CRV: Example II

Q10. Let $X \sim Unif\left(\alpha = -2, \beta = 2\right)$: We define random variable $Y = |X|$ as a function of X. Compute the PDF $f_Y\left(y\right)$ of Y.

A10. Note that $f_X\left(x\right) = \frac{1}{4}$ for $x \in [-2, 2]$

Identifying the Dist. of A Function of A CRV: Example II

Q10. Let $X \sim Unif\left(\alpha = -2, \beta = 2\right)$: We define random variable $Y = |X|$ as a function of X. Compute the PDF $f_Y\left(y\right)$ of Y.

A10. Note that $f_X\left(x\right) = \frac{1}{4}$ for $x \in [-2, 2]$

Step 1: $Y = |X|$ can take values between 0 and 2.

Identifying the Dist. of A Function of A CRV: Example II

Q10. Let $X \sim Unif\left(\alpha = -2, \beta = 2\right)$: We define random variable $Y = |X|$ as a function of X. Compute the PDF $f_Y\left(y\right)$ of Y.

A10. Note that $f_X\left(x\right) = \frac{1}{4}$ for $x \in [-2, 2]$

Step 1: $Y = |X|$ can take values between 0 and 2.

Step 2: Determine the CDF of X. Since $X \sim Unif\left(\alpha = -2, \beta = 2\right)$:

$$F_X\left(x\right) = \frac{x-\alpha}{\beta-\alpha} = \frac{x-(-2)}{2-(-2)} = \frac{x+2}{4}, \quad x \in [-2, 2]$$

Identifying the Dist. of A Function of A CRV: Ex. II (cont'd)

Step 3: In order to determine the PDF of Y, first we need to determine its CDF $F_Y(y)$:

$$F_Y(y) = P\{Y \leq y\} = P\{|X| \leq y\} = P\{-y \leq X \leq y\}$$

which indicates:

$$F_Y(y) = F_X(y) - F_X(-y) = \frac{y+2}{4} - \frac{-y+2}{4} = \frac{y}{2}, \quad y \in [0,2]$$

or in a more detailed form:

$$F_Y(y) = \begin{cases} 0, & y < 0 \\ \frac{y}{2}, & 0 \leq y \leq 2 \\ 1, & y > 2 \end{cases}$$

Identifying the Dist. of A Function of A CRV: Ex. II (cont'd)

Step 4: The PDF of Y can be computed by:

$$f_Y(y) = \frac{\partial F_Y(y)}{\partial y} = \frac{1}{2}, \quad y \in [0,2]$$

or in a more detailed form:

$$F_Y(y) = \begin{cases} 0, & y < 0 \\ \frac{y}{2}, & 0 \leq y \leq 2 \\ 1, & y > 2 \end{cases} \Rightarrow f_Y(y) = \begin{cases} \frac{1}{2}, & 0 \leq y \leq 2 \\ 0, & \text{elsewhere} \end{cases}$$

Identifying the Dist. of A Function of A CRV: Example III

Q11. In a paper plane contest, each contestant will be rewarded by the square of the half of the flight time of their plane. Say, if the flight time in seconds is X, then the reward in dollars is $Y = \frac{X^2}{4}$. The flight time of the planes is exponentially distributed with an average of 5 seconds.

a. Find the PDF of the reward distribution.
b. Find an entrance fee so that the sponsors can break even under a massive participation.

Identifying the Dist. of A Function of A CRV: Ex. III (cont'd)

A11. Note that $f_X(x) = \frac{1}{5}e^{-\frac{1}{5}x}$ for $x \geq 0$

a. **Step 1:** $Y = \frac{X^2}{4}$ can also take non-negative values.

Step 2: Determine the CDF of X. Since $X \sim Exp\left(\lambda = \frac{1}{5}\right)$

$$F_X(x) = 1 - e^{-\frac{1}{5}x}, \quad x \geq 0$$

Step 3: We need to determine the CDF $F_Y(y)$.

$$F_Y(y) = P\{Y \leq y\} = P\left\{\frac{X^2}{4} \leq y\right\} = P\{X^2 \leq 4y\} = P\{-2\sqrt{y} \leq X \leq 2\sqrt{y}\}$$

However, since X is an exponential RV, thus, non-negative:

$$F_Y(y) = P\left\{X \leq 2\sqrt{y}\right\} = F_X\left(2\sqrt{y}\right) = 1 - e^{-\frac{2\sqrt{y}}{5}}, \quad y \geq 0$$

or in a more detailed form:

$$F_Y(y) = \begin{cases} 0, & y < 0 \\ 1 - e^{-\frac{2\sqrt{y}}{5}}, & y \geq 0 \end{cases}$$

Identifying the Dist. of A Function of A CRV: Ex. III (cont'd)

Step 4: The PDF of Y can be computed by:

$$f_Y(y) = \frac{\partial F_Y(y)}{\partial y} = \frac{1}{5\sqrt{y}} e^{-\frac{2\sqrt{y}}{5}}, \quad y \in [0, \infty)$$

$$F_Y(y) = \begin{cases} 0, & y < 0 \\ 1 - e^{-\frac{2\sqrt{y}}{5}}, & y \geq 0 \end{cases} \Rightarrow f_Y(y) = \begin{cases} 0, & y < 0 \\ \frac{1}{5\sqrt{y}} e^{-\frac{2\sqrt{y}}{5}}, & y \geq 0 \end{cases}$$

- This result could have been directly obtained using the theorem of transformation, since $g(x) = \frac{x^2}{4}$ is a strictly monotonic and differentiable in $[0, \infty)$.

Identifying the Dist. of A Function of A CRV: Ex. III (cont'd)

b. The enterance fee should be equal to the expected reward so the sponsors can break even.

$$E[Y] = E\left[\frac{X^2}{4}\right] = \frac{E\left[X^2\right]}{4} = \frac{Var[X] + E[X]^2}{4} = \frac{25+25}{4} = 12.5$$

Note that, for $X \sim Exp\left(\lambda = \frac{1}{5}\right)$

$$E[X] = \frac{1}{\lambda} = 5, \quad Var[X] = \frac{1}{\lambda^2} = 25$$

END OF LECTURE SLIDE 07